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File 1 YOHPAL LIVE — MASTER PRODUCT & ENGINEERING BLUEPRINT

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YOHPAL LIVE — MASTER PRODUCT & ENGINEERING BLUEPRINT

AI-Driven Short Video + Live + Social Commerce Super Platform

1. Core Product Position

YohPal Live should not be a TikTok clone. It should become:

AI-powered short video + live streaming + creator economy + contacts growth + polls + chat + wallet + ads + affiliate platform.

The winning idea:

TikTok entertains.

YohPal Live entertains, sells, pays, connects, votes, teaches, streams, and converts contacts into users.

2. Main Modules

A. Short Video Engine

Core features:

- vertical feed
- instant playback
- AI-ranked For You feed
- Following feed
- Nearby/local feed
- Trending feed
- niche feeds
- offline pre-cache
- video reactions
- comments
- shares
- saves
- duets/stitches
- poll overlays
- shop/booking buttons
- affiliate buttons
- creator profile links

Must-have technical rules:

- all videos must use HLS
- all broken videos must be auto-hidden
- all feeds must prefetch
- all scroll behavior must use predictive loading
- first video must play instantly

Feed architecture:

- Video Upload
 - Transcoding
 - HLS validation
 - Moderation
 - Ranking
 - Feed delivery
 - Engagement tracking
 - Monetisation attribution

B. AI Video & Photo Editing Suite

This is where YohPal can beat TikTok.

Features:

- AI captions
- AI subtitles
- AI translation
- AI background removal
- AI thumbnail generator
- AI hooks generator
- AI title generator
- AI hashtags
- AI music recommendation
- AI voiceover
- AI highlight clipping
- AI auto-trim
- AI beautify
- AI product tagging
- AI scene detection
- AI viral score

Creator workflow:

Upload video

- AI analyses content
- AI suggests edits
- creator applies one-click upgrades
- AI predicts viral score
- creator publishes

Future disruptive features:

- AI clone presenter
- AI avatar video assistant
- AI auto-repurpose to reels/shorts/status
- AI multi-language dubbing
- AI-generated promo posters
- AI sponsor-matching

C. Live Streaming

Core features:

- go live
- live chat
- live gifts
- live polls
- live shopping
- live battles
- live co-hosting
- live moderation
- live recording
- live replay
- live ads

Architecture:

Streamer

- RTMP/WebRTC ingest
- Media server
- HLS/WebRTC playback
- chat engine
- gifts/wallet
- recording
- replay feed

For MVP:

RTMP ingest + HLS playback

For advanced:

- WebRTC interactive live
- multi-host rooms
- creator battles
- low-latency live commerce

D. Multi-Streaming

YohPal should let creators stream once and broadcast everywhere.

Platforms:

YohPal Live
YouTube
Facebook
TikTok
Instagram where API allows
custom RTMP destinations

Flow:

Creator starts live
→ selects destinations
→ YohPal receives stream
→ server forwards RTMP to platforms
→ analytics collected centrally

E. Monetisation Engine

Revenue channels:

ads revenue share
creator gifts
live tipping
paid subscriptions
paid creator channels
boosted videos
brand sponsorships
affiliate earnings
course/product sales
wallet transaction fees
premium editing tools

Creator dashboard:

views
watch time
earnings
gifts
affiliate sales
ad revenue
withdrawals
eligibility score

F. Contacts as User Acquisition Engine

This is a major YohPal advantage.

Features:

phone contacts import
AI invite ranking
family/friends invite flows
WhatsApp share cards
SMS invite fallback
referral tracking
viral squads
creator fan migration

Flow:

User joins
→ imports contacts
→ AI identifies likely joiners
→ generates personalized invite
→ tracks conversion
→ rewards inviter

Important: this must be opt-in and privacy-safe.

G. Search Engine

YohPal Search should be action-based, not just keyword search.

Search across:

videos
creators
live streams
polls

products
services
jobs
courses
communities
hashtags
nearby trends

Search intelligence:

BM25 keyword search
vector semantic search
AI intent detection
ranking by trust + relevance + engagement

Example:

“best electrician near me”
→ Hustle providers + videos + reviews + map

H. Affiliate Marketing Engine

Affiliate should power growth without becoming MLM.

Rules:

earn only from real purchases
earn from active paid accounts
no pay-to-join pyramid
fraud checks required

Affiliate events:

user_signup
course_purchase
video_boost_purchase
subscription_payment
wallet_transaction
market_order

Commission examples:

direct referral: 5–15%
creator product sale: agreed split
course enrolment: recurring active commission

I. Infinity Polls

This can become a viral opinion layer across YohPal.

Features:

poll inside video
poll inside live
poll battles
long-running national polls
sports polls
campus polls
political-safe civic polls
brand polls
creator polls

Poll intelligence:

time-series opinion tracking
demographic segmentation
fraud-resistant voting
AI poll summaries
trend charts
shareable results

J. Ads Engine

Ads must work across the YohPal ecosystem.

Ad types:

in-feed video ads
boosted videos
live stream ads

poll sponsored ads
creator sponsorships
search ads
profile ads
course ads
marketplace ads

Ad targeting:

interest
location
watch behavior
creator category
poll participation
search intent

Billing should be web-based to avoid app store commission issues.

K. Web-Based Wallet

Wallet should power:

creator payouts
gifts
tips
affiliate commissions
ad spend
boosting
subscriptions
marketplace payments
course payments

Rule:

mobile app shows wallet status
web wallet executes payments and withdrawals

This protects against app store payment restrictions.

L. Chat Engine

Chat must connect everything:

creator-to-fan
buyer-to-seller
student-to-lecturer
group chats
live chat
video comments to DM
Hustle client chat
Kitchen order chat
support chat

Architecture:

sequence-based messages
replies
reactions
edits
soft deletes
typing
presence
push notifications
media sharing
AI assistant

3. Main System Architecture

Recommended stack

Flutter mobile app
Firebase Auth
Firestore
Firebase Storage
Cloud Functions
Cloud Run
Redis
PostgreSQL / BigQuery later

OpenSearch later
Media server for live streaming
Web wallet + admin dashboard

4. Main Data Collections

users
creatorProfiles
videos
videoEvents
comments
follows
bookmarks
polls
pollVotes
liveSessions
multistreamSessions
walletSessions
affiliateLinks
affiliateAttributions
adCampaigns
adImpressions
chatConversations
chatMessages
notifications
aiVideoJobs
creatorEarnings

5. Non-Negotiable Engineering Rules

Rule 1: No manual coding outside blueprint

Every developer must follow:

domain model
service
repository
UI adapter
screen
test
rules
deployment

Rule 2: All advanced features must be event-driven

Example:

video_viewed
video_liked
video_shared
creator_followed
poll_voted
gift_sent
wallet_paid
affiliate_converted

Rule 3: Feed must be backend-assisted

Do not depend only on client ranking.

Feed ranking should combine:

watch time
completion rate
rewatches
shares
comments
likes
freshness
creator trust
viewer interests
negative feedback
monetisation priority
safety score

6. MVP Build Order

Phase 1 — Stable Video Foundation

video upload
HLS transcoding
instant feed playback
predictive prefetch
broken video recovery
creator profiles
likes/comments/shares/bookmarks

Phase 2 — AI Creator Studio

AI captions
AI thumbnails
AI hooks
AI hashtags
AI trimming
AI translation
viral score

Phase 3 — Live + Multistream

go live
live chat
gifts
recording
replay
RTMP forwarding

Phase 4 — Growth Engine

contacts import
AI invites
referrals
affiliate attribution
viral share cards

Phase 5 — Monetisation

ads
boosts
creator earnings
wallet
subscriptions
affiliate payouts

Phase 6 — Ecosystem Integration

Hustle
Kitchen
Market
ICS Live
Polls
Search
Chat
Wallet
Ads

7. What Makes YohPal More Disruptive Than TikTok

YohPal should compete through **utility + money + AI**, not only entertainment.

TikTok = watch content

YohPal = watch, earn, sell, hire, learn, vote, chat, stream, and transact

Unique disruptive features:

AI edits every creator into a professional
contacts become acquisition engine
polls turn videos into national conversation
wallet turns engagement into income
affiliate makes every user a growth partner
live commerce connects creators to products
search turns videos into action

8. Developer Reset Instruction

Tell your developer:

Stop patching the old app randomly.

Create a clean YohPal Live v2 branch.

Implement modules in this order:

1. Core video playback and feed stability
2. Creator profile and engagement
3. AI creator studio
4. Live streaming
5. Multistreaming
6. Contacts growth engine
7. Affiliate engine
8. Monetisation and wallet web flow
9. Polls
10. Ads
11. Chat integration
12. Search integration
13. Admin analytics

No feature should be merged unless:

it compiles
it has Firestore rules
it has data models
it has service layer
it has UI adapter
it has test plan

9. Next Smart Move

Build the next document:

YOHPAL LIVE v2 — FULL DEVELOPER EXECUTION ROADMAP

This should define:

folder structure
file-by-file build order
Firestore schema
Flutter services
Firebase Functions
Cloud Run workers
media pipeline
AI workers
wallet web flow
chat integration
ads integration
polls integration
run-to-green checklist

This will stop the manual coding conflict permanently.